# Development of a 5G Core NAS Replay Inspection Tool using UERANSIM*

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**Abstract**

This paper develops an automated tool to verify 5G NAS replay vulnerabilities and evaluates it using an Open5GS testbed. The AMF successfully rejected duplicate messages, demonstrating effective automated security compliance auditing.

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# 1 Introduction

To meet KISA's 5G private network guidelines for preventing NAS message replay and algorithm downgrade attacks during Security Mode negotiation, we developed an automated inspection tool. By modifying UERANSIM to inject duplicate NAS Security Mode Complete messages, we validated the network's security efficacy on an Open5GS testbed.

# 2 Proposed Method

## 2.1 Inspection Objectives and Test Scenario

The proposed tool evaluates 5GC compliance with KISA's replay protection requirements [2]. Focusing on the UE-AMF NAS security mode negotiation, it injects duplicate Security Mode Complete messages following a Security Mode Command to test the core network's exception-handling capability against replay anomalies.

## 2.2 Tool Architecture and Source Code Modification

To implement the tool, we analyzed and modified the C++ source code of UERANSIM, an open-source 5G UE/RAN simulator [4]. We modified the transmission logic within the NAS layer module responsible for security mode responses. Specifically, we duplicated the function call sendNasMessage(resp); to execute twice in immediate succession. This forces the simulator to transmit two identical response messages within a sub-millisecond interval, acting as a protocol-level replay injection tool.

# 3 Results and Evaluation

## 3.1 Testbed Environment

To evaluate the developed tool, a virtual 5G network testbed was constructed on a Linux environment. The core network was deployed using Open5GS [5], and the replay injection module was executed via the modified UERANSIM [4]. Both components were interconnected locally using standard loopback interface parameters.

## 3.2 Inspection Results and Analysis

4 Upon execution, the modified simulator transmitted duplicate Security Mode Complete messages during the NAS security negotiation phase. Packet analysis via Wireshark confirmed that the second duplicate message was injected approximately 0.00004 seconds after the initial valid message. The Open5GS AMF accepted the first message while discarding the subsequent duplicate, preventing protocol ambiguities.

## 3.3 Security Verdict

Despite the signaling replay injection, the core network maintained its protocol state machine stability. The UE completed the registration procedure without service disruption, activating the virtual interface (uesimtun1) with a valid IP address (10.45.0.3). Consequently, the testbed demonstrated satisfactory compliance (PASS) with KISA security requirements [2].

# 3 Conclusion

This paper proposed a protocol-level automated inspection tool using a modified UERANSIM to verify NAS signaling replay protection. Evaluation on an Open5GS testbed confirmed that the core network effectively discarded duplicate security messages within 0.00004 seconds, successfully establishing the virtual interface (uesimtun1) without service disruption. This demonstrates compliance with KISA guidelines [2], proving the tool's utility as a pre-deployment compliance audit framework to prevent downgrade attacks in private 5G networks.

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# References

[1] 3GPP. (2021). *System architecture for the 5G System (5GS)* (3GPP TS 23.501, Version 17.0.0). https://www.3gpp.org/specifications-technologies/standards
[2] Korea internet & security agency. (2026, March). *5G Private Network Security Guide*.
[3] 3GPP. (2021). *Security architecture and procedures for 5G system* (3GPP TS 33.501, Version 17.1.0). https://www.3gpp.org/specifications-technologies/standards
[4] Aligungr. (n.d.). UERANSIM: Open source 5G UE and RAN simulator [Source code]. GitHub. https://github.com/aligungr/UERANSIM
[5] Open5GS. (n.d.). Open5GS: Open source project for 5G Core and EPC [Source code]. GitHub. https://github.com/open5gs/open5gs